

# Auto Analog RL

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**AUTOMATED SIZING OF HIGH SPEED INTERFACE  
CIRCUITS VIA REINFORCEMENT LEARNING**

**TEAM BELUGA**

Bharat Kumar Saxena

Medha Soni

Samarth Soni



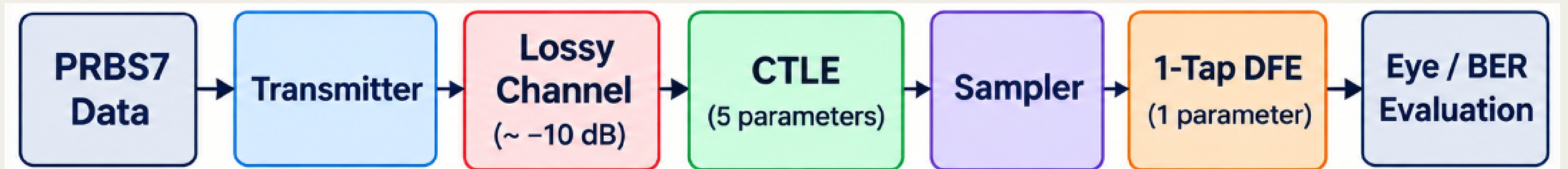
# Why automate analog circuit sizing?

- Analog circuit performance depends on multiple coupled design parameters
- Changing one parameter can affect gain, bandwidth, power, linearity and eye opening
- Finding a design that satisfies all specifications simultaneously requires repeated SPICE simulations
- Manual sweeps become expensive as the design space and number of constraints increase



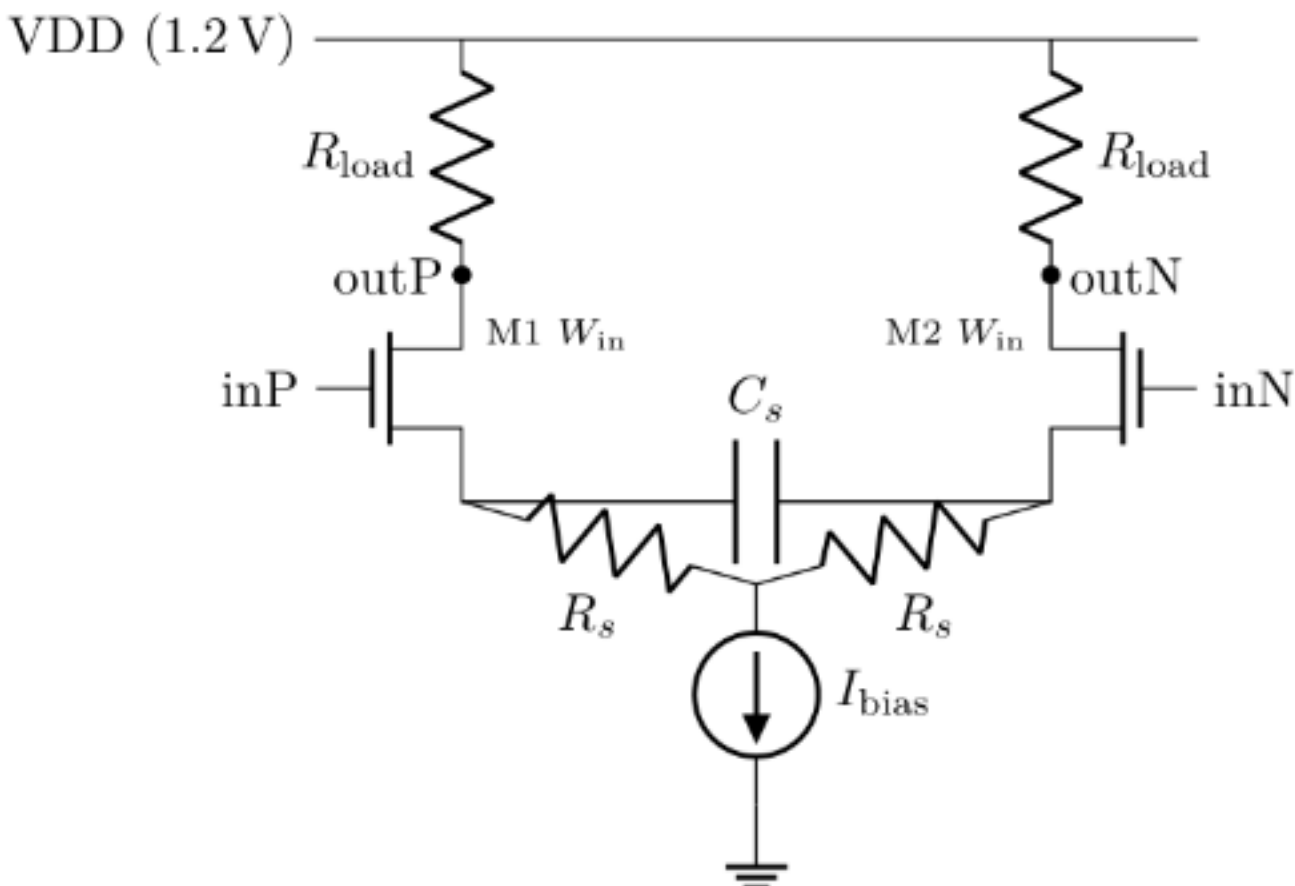
# PCIE RECEIVER: CTLE + DFE

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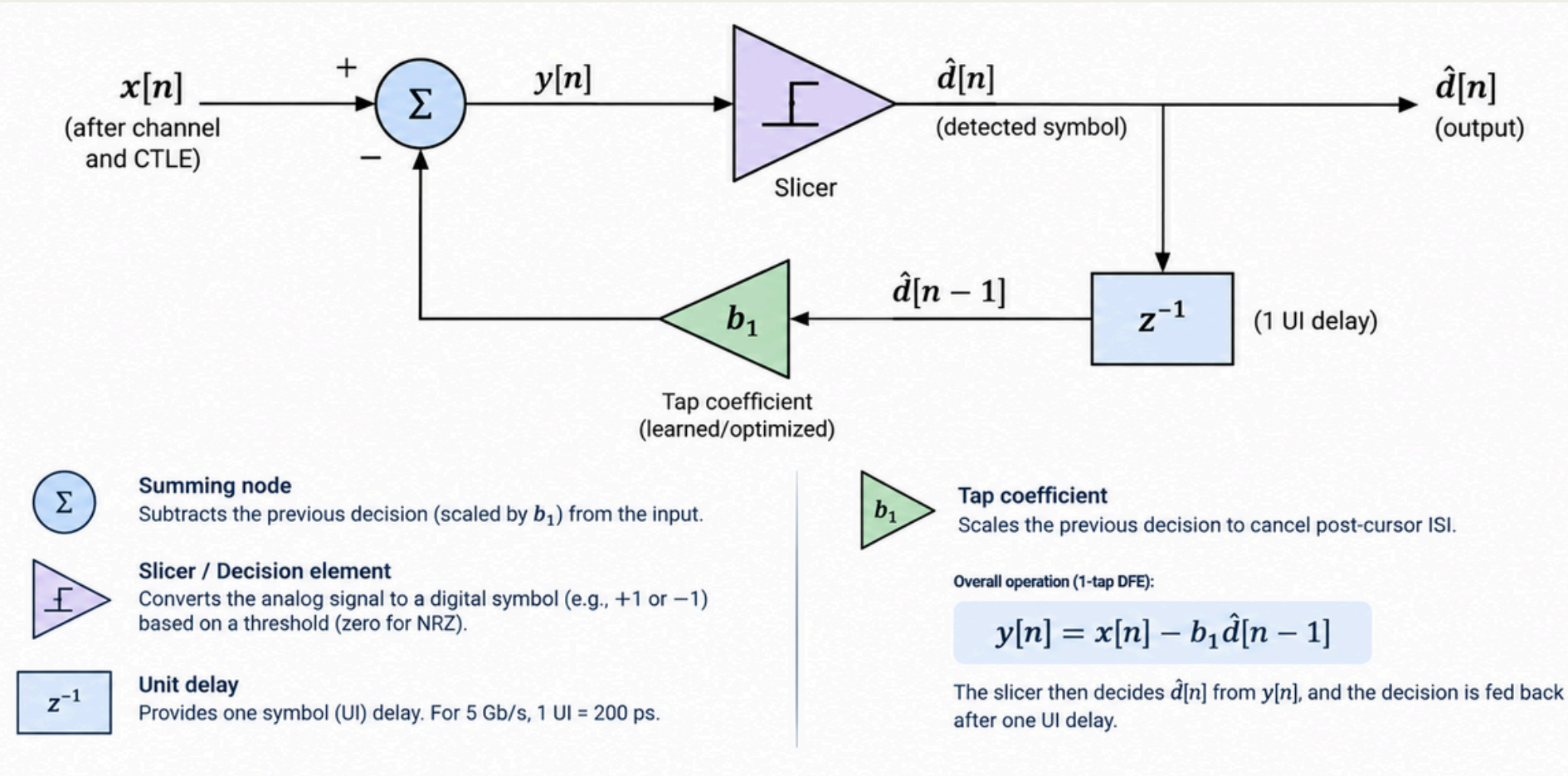


- Channel: High-frequency loss + Inter Symbol Interference(ISI)
- CTLE: Compensates frequency-dependent loss
- DFE: Removes residual ISI using the previous decision

# CTLE AND DFE CIRCUIT



One-stage CTLE with source degeneration



One – Tap DFE



# WHAT ARE WE OPTIMIZING?

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## Design Variables

$W_{in}$  |  $R_{load}$  |  $I_{bias}$  |  $R_s$  |  $C_s$  | DFE tap

## Performance Constraints:

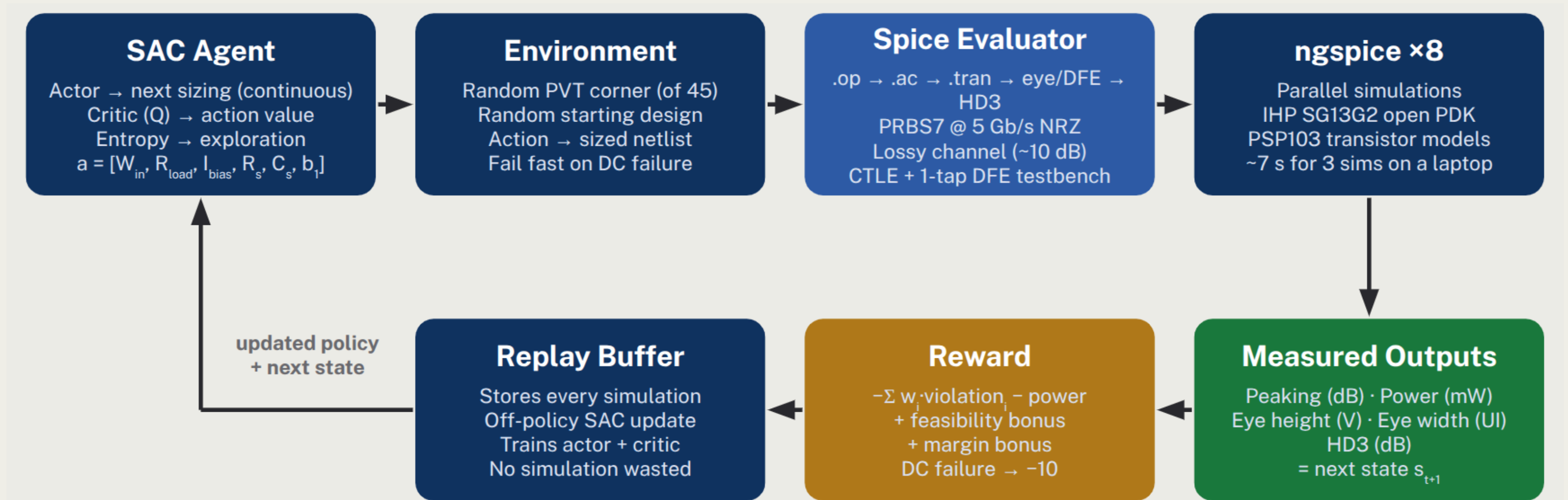
| Specification | Meaning                       | Our Target   |
|---------------|-------------------------------|--------------|
| Eye Height    | Voltage margin                | $\geq 0.25V$ |
| Eye Width     | Timing margin                 | $\geq 0.7UI$ |
| Peaking       | HF boost                      | 3-12dB       |
| Power         | Energy/Consumption constraint | $\leq 2mW$   |
| HD3           | Linearity/Distortion          | $\leq -30dB$ |

A DESIGN IS FEASIBLE ONLY WHEN ALL CONSTRAINTS ARE SATISFIED



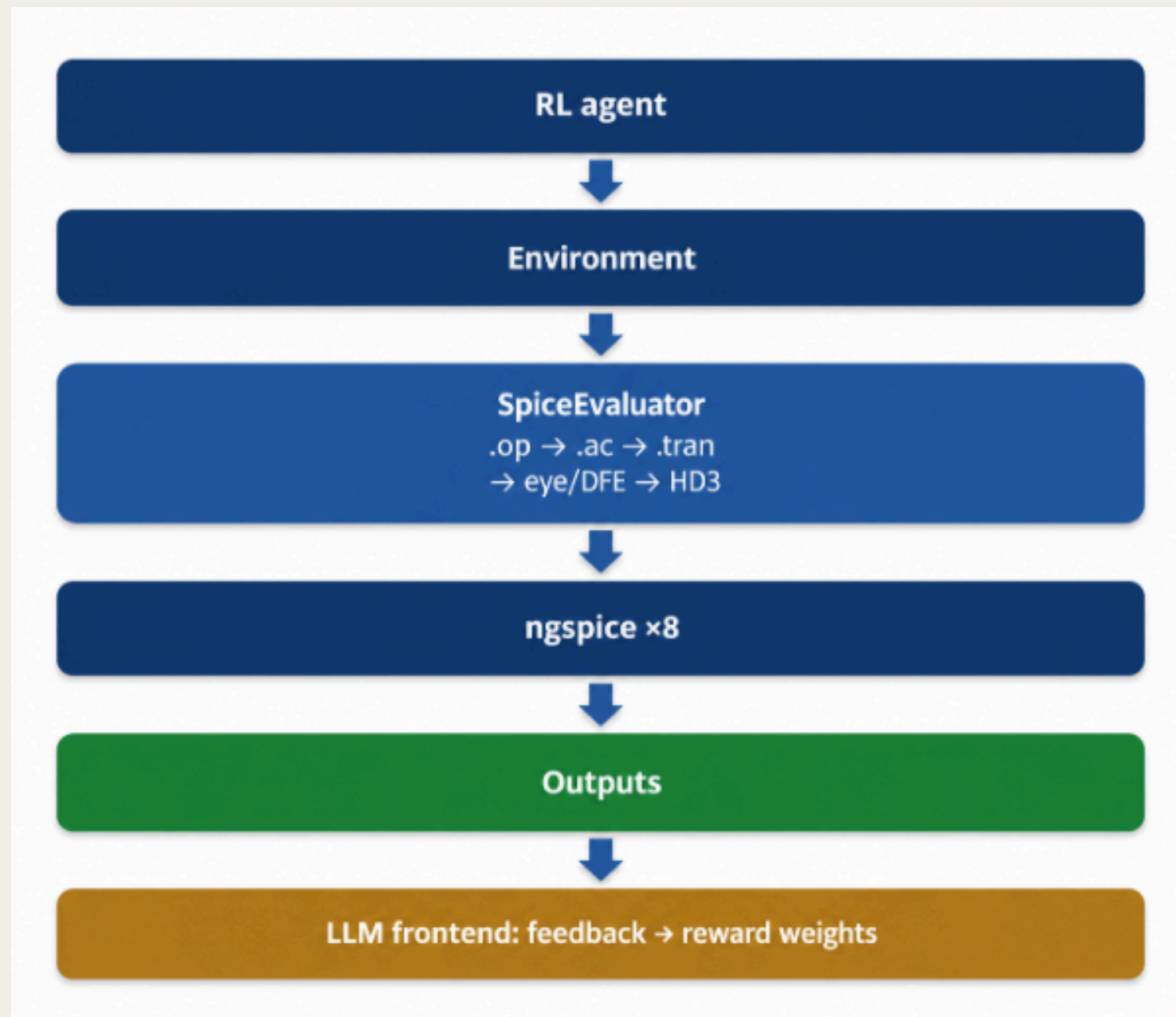


# ARCHITECTURE





# CIRCUIT SIZING AS AN RL PROBLEM



## State

- Circuit performance from previous simulation

## Action

- $[W_{in}, R_{load}, I_{bias}, R_s, C_s, DFE\ Tap]$

## Reward

- Based on:
- Specification violations
- Power
- Performance margin
- Feasibility bonus



# SOFT ACTOR CRITIC (SAC)

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## Why SAC?

- Soft Actor-Critic is an off-policy reinforcement learning algorithm designed for continuous action spaces.

## Our application

- The SAC agent learns to select circuit parameters that maximize reward while satisfying design specifications.

- **Actor** → generates circuit parameters
- **Critic** → evaluates action quality
- **Entropy** → encourages exploration
- **Continuous action space** → suitable for analog sizing





# TRAINING THE SAC LOOP

```
• AutoAnalog-RL policy rollout model=sac_final.zip specs: eye >= 0.25 V, width >= 0.7 UI, peaking 3-12 dB, power <= 2 mW, HD3 <= -30 dB
• start: random design (seed 21), corner SS_1.00V_62.5C

• step    W_in  R_load  I_bias   R_s    C_s  peak dB power mW  eye V eye UI  HD3 dB  reward  specs
• -----
•   1    27.62u   829    0.46m   307    885f   5.25   0.55  0.173  0.66    -   -1.64 OK:peak OK:peak<12 OK:power --:width --:eye --:hd3
•   2    24.05u   912    0.65m   263    786f   5.17   0.78  0.246  0.76    -   -1.41 OK:peak OK:peak<12 OK:power OK:width --:eye --:hd3
•   3    23.24u   977    0.84m   232    688f   4.97   1.01  0.321  0.80  -57.8  20.21 OK:peak OK:peak<12 OK:power OK:width OK:eye OK:hd3

• all specs met after 3 simulation step(s) in 7.0 s of wall time
• sized netlist header:
• * AutoAnalog-RL sized CTLE
```

**3 sims**

to meet every spec

**7 s**

wall time on a laptop

**SS / 1.0 V / 62 °C**

random corner, random start

Reward =  $-\sum w_i \cdot \text{violation}_i$  - power  
charge + success bonus + margin bonus  
DC failure: -10, fail fast

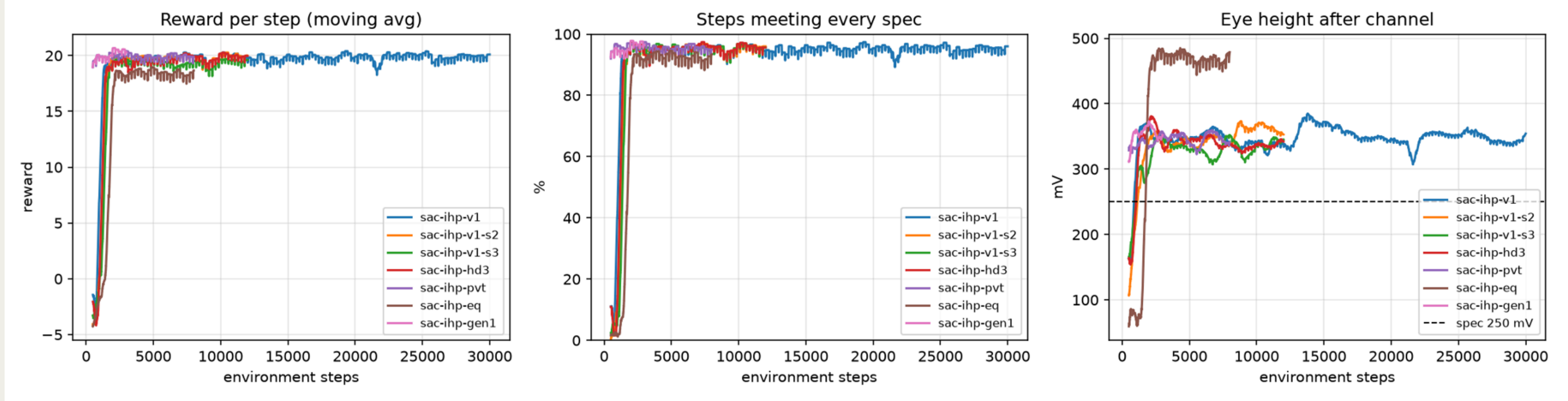


# Live Demo



# TRAINING AND CONVERGENCE (THREE SEEDS)

SAC training on IHP sg13g2 PSP103 models (window 500 steps)



**~2k steps**

to converge, every run

**60 / 60**

rollouts feasible, 3 seeds

**$2.53 \pm 0.06$**

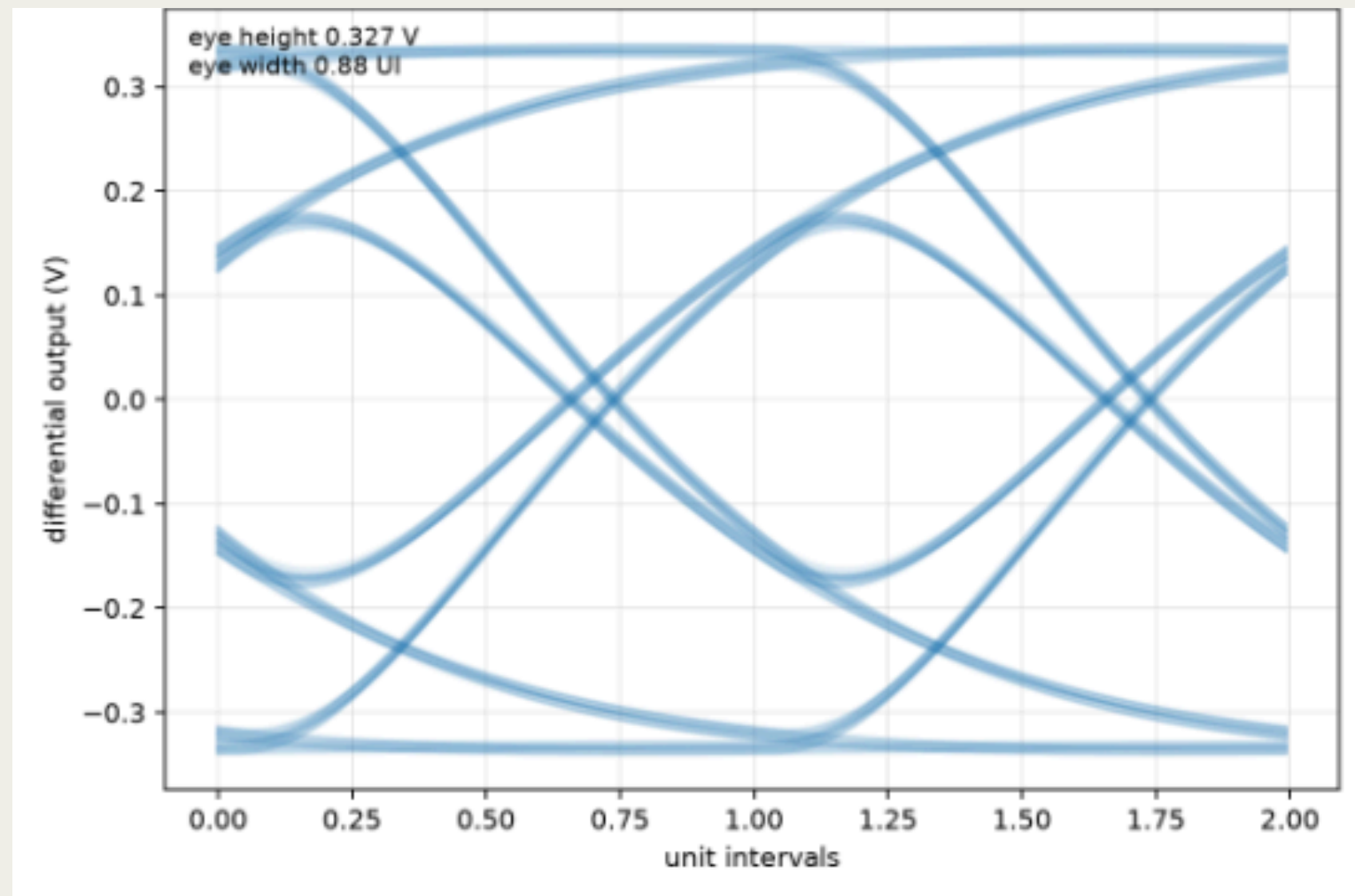
steps to feasibility (mean over seeds)

**95 %**

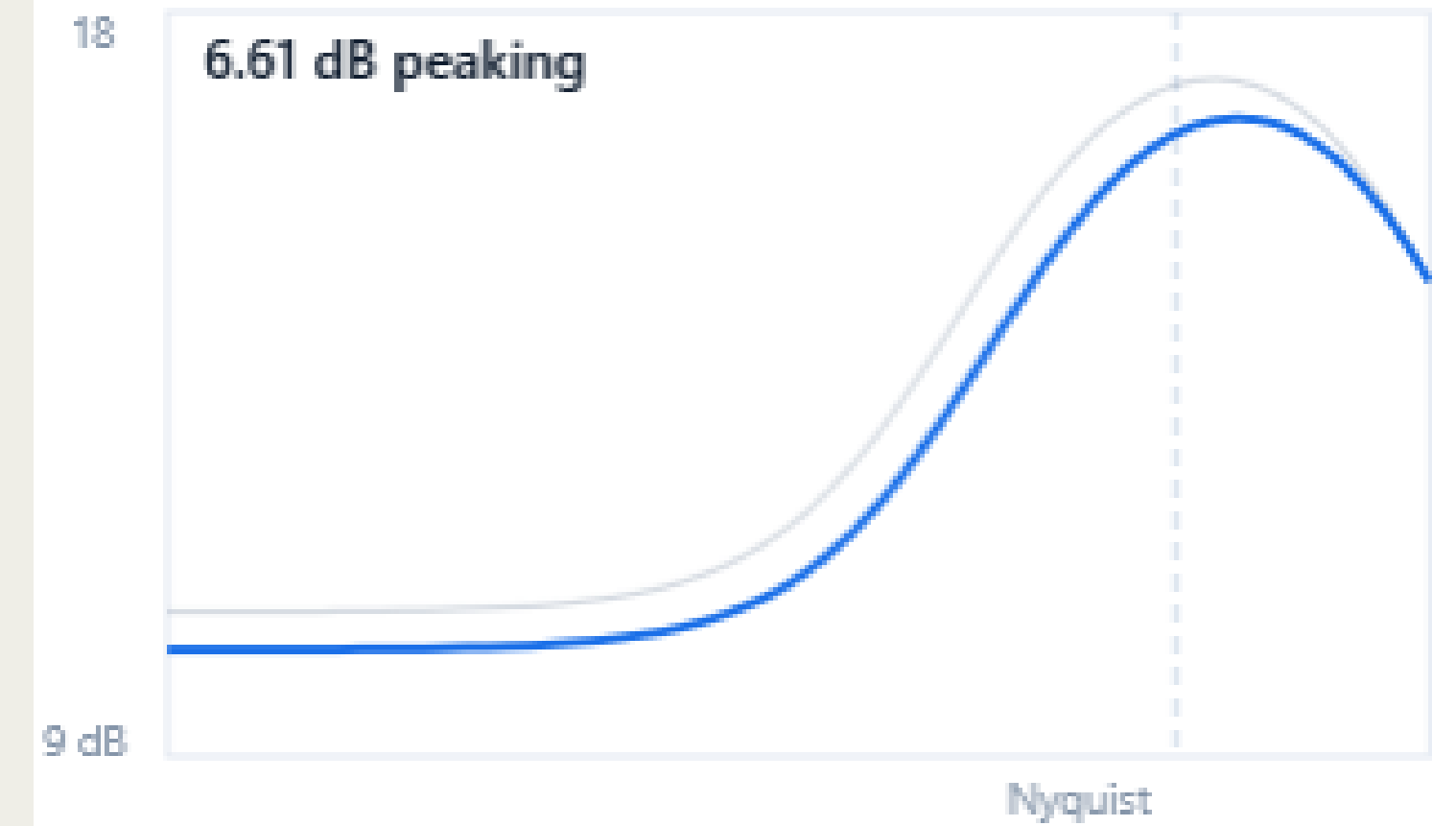
of training steps meet every spec



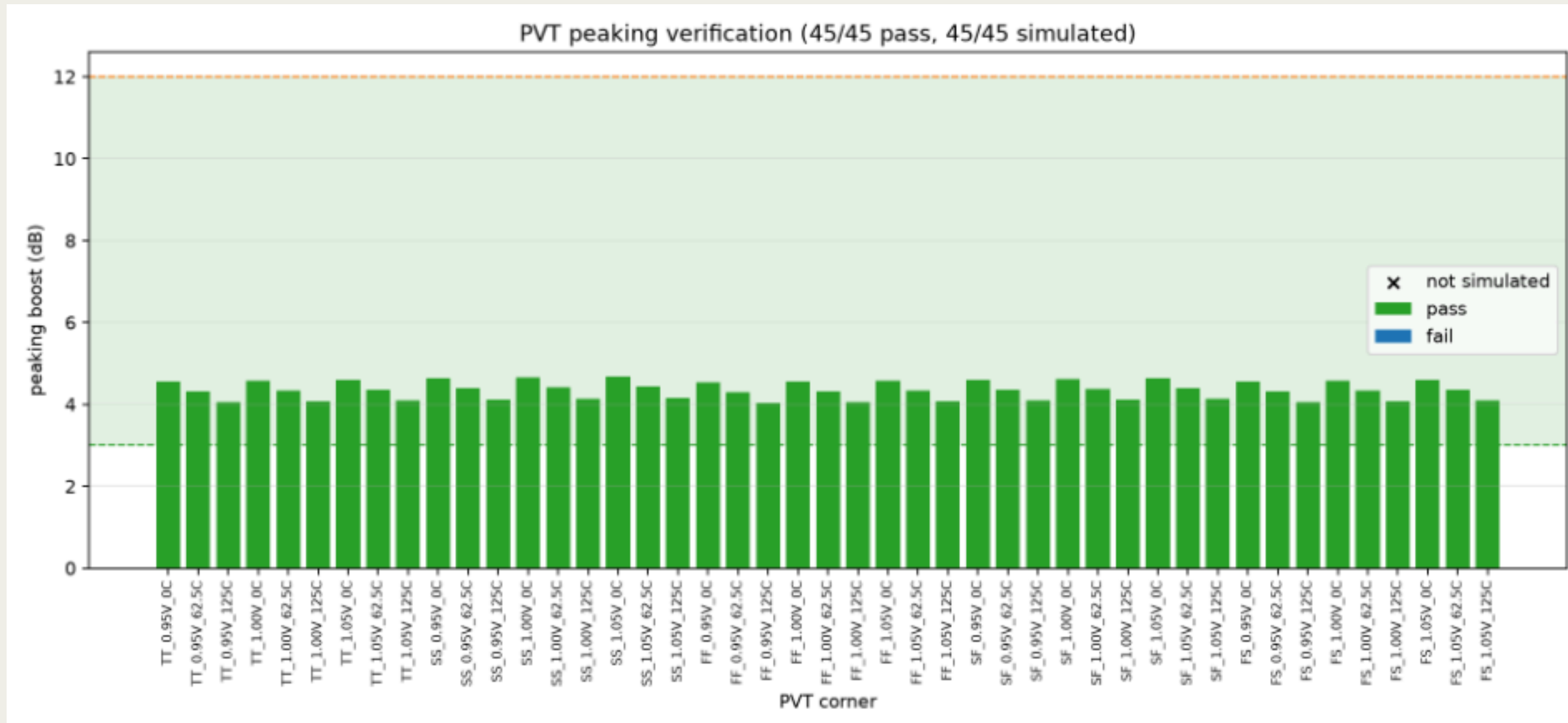
# EYE DIAGRAM + AC RESPONSE



## FREQUENCY RESPONSE



# PVT ROBUSTNESS



## Results

45 / 45 corners pass  
24 / 24 random-corner  
rollouts feasible  
Best curriculum-  
trained result:

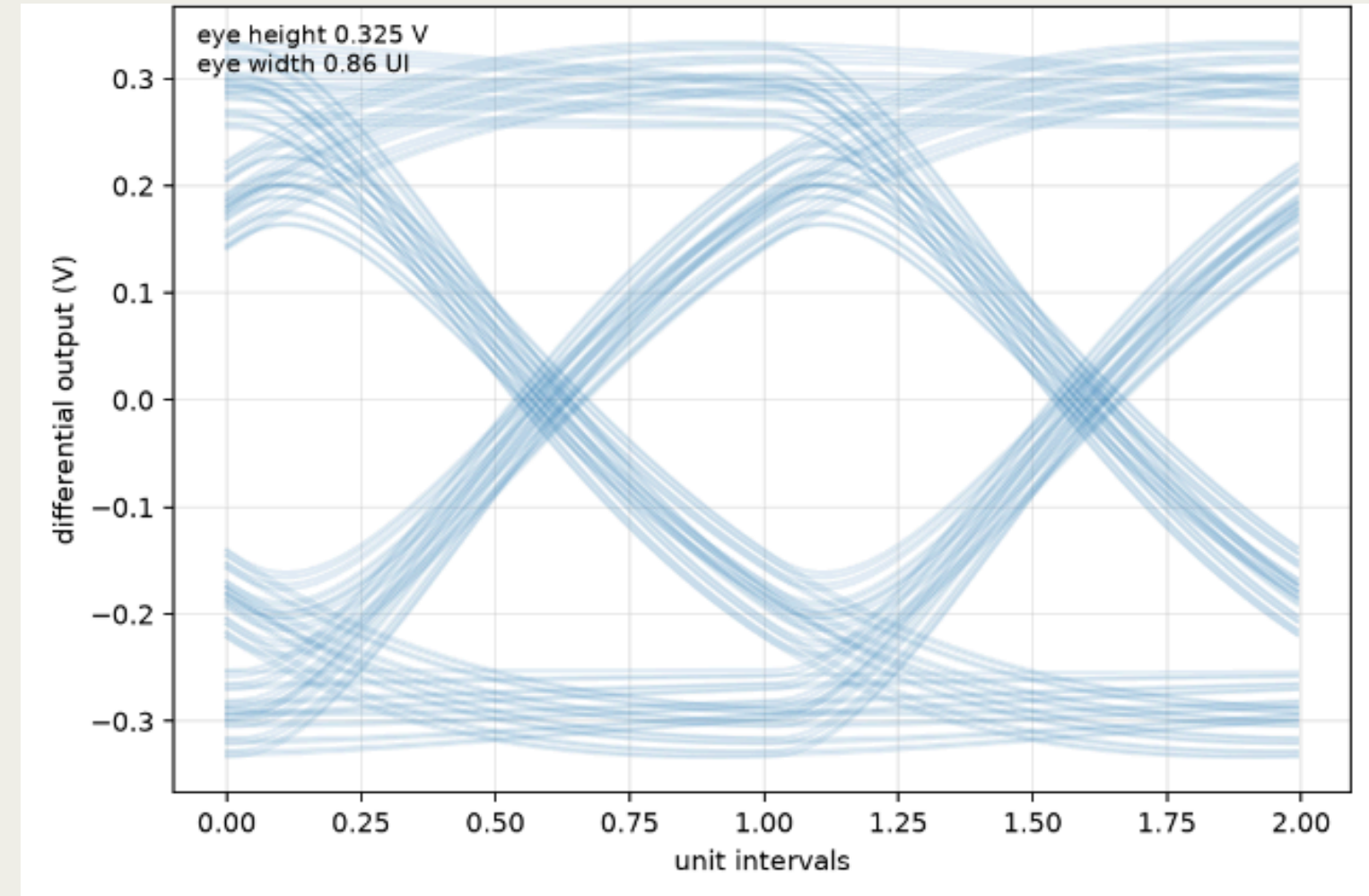
- Eye height: 0.370 V
- Power: 1.12 mW
- HD3: -61.9 dB





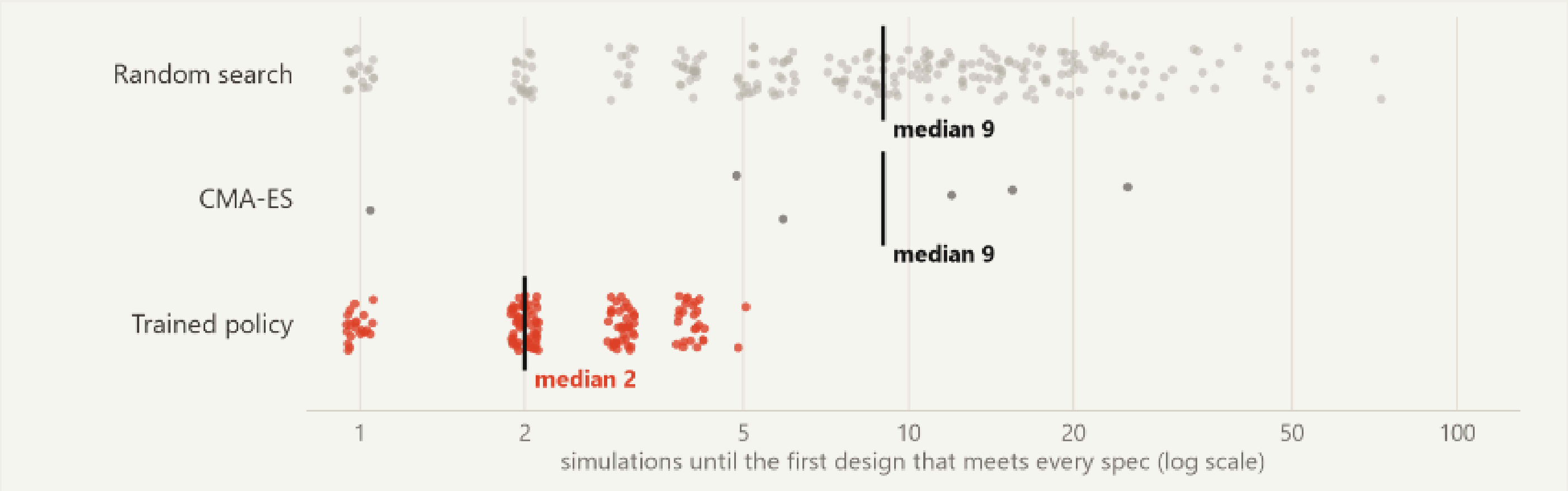
# DFE EXTENSION : COMPLETE EQUALIZER

Extending the action space to six values places the DFE tap b1 under the same policy as the CTLE parameters, with the post-DFE eye height driving the reward. And the optimisation converges on a balanced allocation rather than loading either stage.





# COMPARISON WITH CONVENTIONAL SEARCH



| Method (random starting designs) | First feasible design | Per new instance               |
|----------------------------------|-----------------------|--------------------------------|
| SAC policy (144 rollouts)        | median 2–3, worst 5   | 2–3 simulations, no search     |
| CMA-ES, random start (6 trials)  | median 9, worst 24    | ~10 to feasible; 400 to refine |
| Random search (3000 designs)     | #27 (7.9 % feasible)  | ~13 per feasible design        |
| AC-only bounded search (pre-ML)  | never reaches the eye | 0.160 V post-channel eye       |



# SUMMARISING WHAT WE ACHIEVED

| Specification            | Requirement                                   | Measured                                        | Status |
|--------------------------|-----------------------------------------------|-------------------------------------------------|--------|
| Nyquist frequency        | 2.5 GHz                                       | PRBS7 at 5 Gbit s <sup>-1</sup>                 | Pass   |
| High-frequency peaking   | 3–12 dB                                       | 4.5 dB at 2.5 GHz                               | Pass   |
| Peaking tunability       | 1.25–2.5 GHz                                  | 5.23 dB at 1.25 GHz (retuned)                   | Pass   |
| Topology                 | 1-stage CTLE, variable $R_s, C_s$ + 1-tap DFE | as sized, NRZ PRBS7                             | Pass   |
| Eye height after channel | $\geq 100$ mV                                 | 370 mV                                          | Pass   |
| Eye width after channel  | $\geq 0.4$ UI                                 | 0.90 UI                                         | Pass   |
| Linearity (HD3)          | $\leq -30$ dB                                 | -61.9 dB                                        | Pass   |
| Input-referred noise     | $\leq 1.5$ mV rms                             | 0.242 mV rms                                    | Pass   |
| Power                    | $\leq 15$ mW                                  | 1.12 mW                                         | Pass   |
| Active area              | $\leq 0.05$ mm <sup>2</sup>                   | $1.1 \times 10^{-5}$ mm <sup>2</sup> (estimate) | Pass   |
| PVT coverage             | 45 corners                                    | 45/45                                           | Pass   |



# Thank you!

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